

A Multimodal Approach to Music Emotion Analysis Integrating Audio and Lyrics

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Abstract

This study addresses music emotion recognition (MER) by predicting valence and arousal from both audio and lyrics using pretrained audio spectrogram transformer (AST) and bidirectional encoder representations from transformers (BERT) models. Our models are fine-tuned on the MERGE dataset and evaluated on PMemo, our multimodal approach significantly outperforms unimodal baseline models. Among tested architectures, the cross-attention model achieves the best performance. Lyrics contribute more to valence prediction, while audio is more effective for arousal. We also note that emotion-focused similarity may reduce recommendation diversity, highlighting the need to balance emotional accuracy with variation. These findings support the use of multimodal models for more human-aligned and nuanced music emotion understanding.

1 Introduction

Music emotion recognition (MER) aims to accurately identify the emotional content conveyed by songs, which inherently integrate both audio and lyrical modalities. This study proposes a multimodal regression approach to predict valence and arousal by combining audio and lyrical features using pretrained Audio Spectrogram Transformer (AST) and Bidirectional Encoder Representations from Transformers (BERT) models. Our models are fine-tuned on the MERGE dataset and evaluated on both the MERGE and PMemo datasets. Experimental results show that our multimodal method significantly outperforms single-modal baseline models, with the cross-attention architecture achieving the highest performance across both valence and arousal predictions. We further observe that lyrics predominantly contribute to valence prediction, while audio features offer stronger cues for arousal. These findings demonstrate the effectiveness of multimodal models in achieving a deeper and more accurate understanding of musical emotions. However, we also identify a key limitation: the model tends to underestimate valence intensity compared to human perception, indicating the need for adjustment strategies to better align predictions with subjective emotional experience.

2 Data Description

2.1 Music Emotion Recognition Generative Emotion dataset

For this project, we utilized the MERGE dataset, which is specifically designed for Music Emotion Recognition research [1]. The MERGE dataset is a comprehensive collection that includes both

audio recordings and lyrics. It is publicly available through Zenodo [2].

2.1.1 Dataset Composition. MERGE dataset offers several variants to accommodate different needs. Audio components provide 30-second excerpts for each music sample, while the lyrical components include the entire lyrics for each song. The dataset is available in both complete and balanced versions, with the following sample distributions:

- MERGE Audio Complete: 3,554 samples
- MERGE Audio Balanced: 3,232 samples
- MERGE Lyrics Complete: 2,568 samples
- MERGE Lyrics Balanced: 2,400 samples
- MERGE Bimodal Complete: 2,216 samples
- MERGE Bimodal Balanced: 2,000 samples

The MERGE Bimodal Complete dataset exhibits an imbalanced distribution across the emotional quadrants: Q2 has the most samples (673), followed by Q1 (525), Q4 (518), and Q3 (500). Although this imbalance reflects natural emotional variation, it may introduce bias during training. Therefore, we primarily used the balanced subset to ensure fair model evaluation, while using the complete version only for supplementary analysis.

2.1.2 Dataset Selection and Rationale. We mainly utilized the MERGE Bimodal Balanced dataset, which contains 2,000 samples with both audio and lyrics data. This balanced version ensures equal representation across the four quadrants of the Valence-Arousal emotional space, helping to reduce bias in model training and evaluation. We chose this version to ensure a more evenly distributed emotional representation.

Additionally, we conducted supplementary experiments with the MERGE Bimodal Complete dataset (2,216 samples) to assess our model’s performance under slightly imbalanced conditions. This allowed us to evaluate how well our approach generalizes to real-world scenarios where certain emotional categories may be overrepresented [2].

2.1.3 Annotation Schema and Emotional Representation. The dataset provides emotion annotations based on the Valence-Arousal space, following a variant of Russell’s circumplex model of affect [3]. In the original model, valence and arousal values typically range from -1 to 1, but the MERGE dataset normalizes them to a 0–1 scale. In this setting, a neutral threshold is set at 0.5 for both valence and arousal.

Each sample is labeled with continuous values for both valence (emotional positivity) and arousal (emotional intensity). Additionally, these continuous values are mapped to four quadrants (Q1–Q4) representing general emotional regions:

- Q1 (high valence, high arousal): emotions such as happiness or excitement
- Q2 (low valence, high arousal): emotions such as anger or fear
- Q3 (low valence, low arousal): emotions such as sadness or depression
- Q4 (high valence, low arousal): emotions such as calmness or relaxation

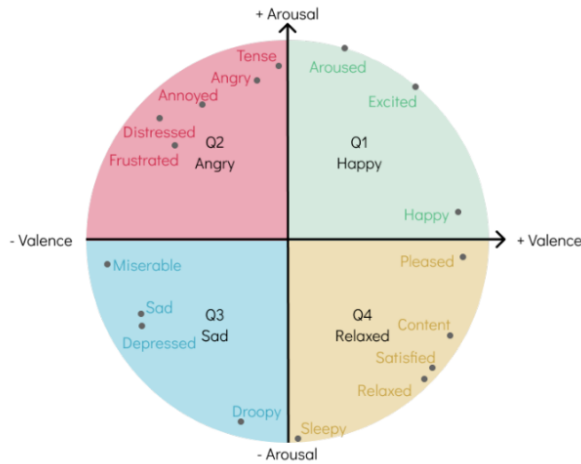


Figure 1: MERGE dataset mapped onto Russell’s circumplex model of emotion [2].

It should be noted that these emotion labels are illustrative examples, and specific emotional states within each quadrant may vary.

2.1.4 Data Usage. During our fine-tuning process, we incorporated both audio features and lyrical content as inputs to our model. The quadrant information was not used during training but served as a categorical reference for calculating classification metrics such as F1 scores and accuracy, allowing us to evaluate our model’s performance beyond simple regression metrics.

For training, testing, and validation, we used the pre-split TVT dataframes (70/15/15 ratio) provided by the MERGE dataset without any modifications. This split ensures consistency and reproducibility of our results, while providing sufficient data for model training and evaluation.

2.2 Perceptual and Personal Emotion in Music

To complement our evaluation, we utilized the PMemo dataset—a publicly available resource designed for music emotion recognition. It consists of 642 filtered songs, each annotated with mean valence and arousal scores based on a 40-second chorus segment. We used only the audio and lyrics modalities, applying the standard Train/Validation/Test split (461/90/91). Although the dataset

also includes physiological signals (EDA), they were excluded from our study. PMemo’s multimodal design and consistent annotation schema make it a strong benchmark for evaluating emotion prediction models [4, 5].

3 Methodology

3.1 Data Preprocessing

3.1.1 Audio Data Preprocessing. The audio data consists of 30-second MP3 song clips. All files are resampled to 16 kHz to match the AST pretraining setup, and stereo signals are converted to mono by averaging channels. The audio is then normalized to zero mean and unit variance. We use the AST Feature Extractor, which transforms the audio into log-Mel spectrograms suitable for deep learning [6, 7].

3.1.2 Text Data Preprocessing. The lyrics data is provided in plain text format (.txt files) and has already undergone basic preprocessing in the MERGE dataset, so no additional cleaning is necessary. The text is directly tokenized using a pretrained BERT tokenizer, which splits the text into subword units while preserving semantic meaning. Sequences are automatically truncated or padded to a fixed maximum length of 128 tokens to maintain uniform input dimensions [8, 9].

3.2 Baseline Model

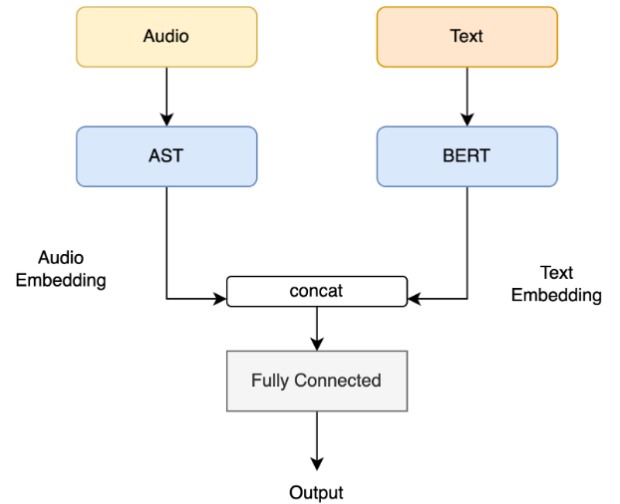


Figure 2: Baseline Architecture: Concatenation of AST and BERT embeddings

3.2.1 Baseline Architecture. Our baseline architecture integrates audio and text features through early concatenation to predict emotional content in music. This design choice was motivated by the need to establish a straightforward but effective foundation against which we could compare more sophisticated fusion techniques. Figure 2 illustrates the model architecture with the following specifications:

- **Audio Encoder:** We utilize a pretrained AST model (MIT/ast-finetuned-audioset-10-10-0.4593) [6, 7]. The input audio is first converted into log-Mel spectrograms before being processed by AST to generate fixed-dimensional embeddings.
- **Lyrics Encoder:** We employ a pretrained BERT-base model (bert-base-uncased) [8, 9], where input lyrics are tokenized and embedded into contextual representations.
- **Feature Fusion:** The audio and text embeddings are concatenated along the feature dimension to form a joint representation.
- **Projection Layer:** A fully connected layer maps the fused representation into a common latent space.
- **Regression Head:** A final multilayer perceptron (MLP) with one hidden layer predicts two continuous outputs: valence and arousal scores.
- **Fusion Method:** Early concatenation combines the 768-dimensional audio and text embeddings into a single 1536-dimensional feature vector. This straightforward design enables us to establish a clear and interpretable benchmark, against which we can evaluate the effectiveness of more sophisticated fusion techniques.
- **Loss Function:** We use Mean Squared Error (MSE) loss to train the model, minimizing the difference between predicted and true emotional scores.

3.2.2 Training Configuration.

- **Batch Size:** A batch size of 16 was determined through experimental tuning as a balance between computational efficiency and model convergence stability.
- **Learning Rate:** A learning rate of 2×10^{-5} was selected after empirical testing to ensure stable convergence without overshooting minima in the loss landscape.
- **Epochs:** Training for 5 to 6 epochs was sufficient for convergence without overfitting, as validated through performance on the validation set.
- **Computational Resources:** We primarily utilized Google Colab (GPU A100) and RunPod due to their accessibility and sufficient computational power for our modeling needs.
- **Optimizer:** We used the Adam optimizer for its adaptive learning rate capabilities, which facilitate faster and more stable convergence in deep learning tasks.

3.3 Ablated Systems

To rigorously evaluate the contribution of different modalities and fusion techniques, we implemented several ablation studies. These variations help us understand which components contribute most significantly to emotion prediction performance.

3.3.1 Singlemodal Systems.

- **AST only:** This model uses only the audio transformer to establish the performance ceiling for audio-based emotion prediction, quantifying the contribution of audio signals to the overall emotional assessment.
- **BERT only:** This model uses only the text transformer to assess how much emotional information is conveyed through lyrics alone.

- **VGGish only:** This model uses a CNN-based audio feature extractor to compare the effectiveness of convolutional architectures against transformer-based ones for emotion prediction.

3.3.2 Multimodal Systems.

- **Cross Attention:** This fusion method enables one modality (e.g., text) to selectively attend to features in another modality (e.g., audio). It is designed to capture complementary emotional cues that may not be evident in individual modalities.
- **Cross Attention + Half Training Dataset:** We evaluated whether cross-attention models can maintain performance when trained on only 50% of the available data, to assess data efficiency and robustness under constrained settings.
- **Modality Attention:** This mechanism dynamically adjusts the weighting between audio and text inputs, assigning more importance to the modality that carries stronger emotional signals in a given sample.
- **Gated Fusion:** A gating mechanism controls the contribution of each modality at the final stage of prediction. This was motivated by the idea that simple concatenation may not effectively handle contradictory emotional signals between modalities.

3.4 Evaluation Metrics

To comprehensively evaluate model performance across different dimensions of emotion prediction, we utilized multiple complementary metrics.

3.4.1 Regression Metrics.

- **MSE (Mean Squared Error):** Measures the average squared difference between predictions and true values. We used MSE as the primary training loss to align with prior work in emotion regression tasks and ensure a stable optimization objective.
- **MAE (Mean Absolute Error):** Calculates the average magnitude of prediction errors without considering their direction, providing an interpretable measure on the original 0–1 scale for both valence and arousal.
- **CCC (Concordance Correlation Coefficient):** Assesses the agreement between predicted and true values by considering both correlation and mean-variance differences (range: $[-1, 1]$). We prioritized CCC during evaluation because it captures both linear correlation and prediction bias, offering a more comprehensive view of prediction quality.

$$CCC = \frac{2\rho\sigma_y\sigma_{\hat{y}}}{\sigma_y^2 + \sigma_{\hat{y}}^2 + (\mu_y - \mu_{\hat{y}})^2}, \quad (1)$$

where μ is mean, σ is standard deviation, and ρ is Pearson correlation coefficient.

- **R² (Coefficient of Determination):** Measures the proportion of variance in the true values explained by the model (range: $(-\infty, 1]$). Although R² and CCC are conceptually related, we included both metrics to better understand model performance from multiple statistical perspectives.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (2)$$

3.4.2 Classification Metrics.

- **F1 Scores:** Calculated for each quadrant (Q1–Q4) in the valence-arousal space, enabling us to evaluate performance across different emotion categories (e.g., happy, angry, sad, calm).
- **Accuracy:** Overall classification accuracy across quadrants provides a single summary measure of quadrant prediction performance.

We included both regression and classification metrics to provide a complete picture of performance, as some applications might prioritize continuous emotion values while others might focus on discrete emotion categories.

3.5 Modern Song Evaluation

To assess real-world generalization capability beyond the training datasets, we evaluated our models on ten modern songs not included in the training data. Notably, the original training data is heavily skewed toward songs from the 1950s to early 2000s. In contrast, the newly selected songs are from 2000 to 2025, representing more contemporary musical styles. This evaluation was conducted to examine whether the models can handle differences in musical trends and stylistic shifts across eras.

3.5.1 Annotation Process.

- **Human Raters:** Each song was annotated by three human raters to establish a more reliable ground truth, accounting for the subjective nature of emotional perception.
- **Annotation Protocol:** Ratings included valence (0–1), arousal (0–1), and quadrant assignment based on threshold values of 0.5.
- **Consensus Calculation:** Average scores were calculated to create a human consensus, reducing individual bias and providing a more stable reference point.

3.5.2 Model Comparison.

- **Multiple Model Evaluation:** We compared the performance of the cross-attention, gated fusion, and modality attention models against the human-annotated consensus to identify which approach best aligns with perceived emotional content.
- **External Baseline:** We also included GPT-4o as a reference model to explore how general-purpose language models perform.
- **Visualization Approach:** Performance was visualized in valence-arousal space to identify patterns in model predictions and areas where models struggle with certain types of emotional content.

3.6 Extension Studies

3.6.1 Cross-Modal Retrieval. When conducting multimodal music emotion analysis, we assume that audio and lyrics share an underlying feature space. Building on the work of Yu [10], who used a

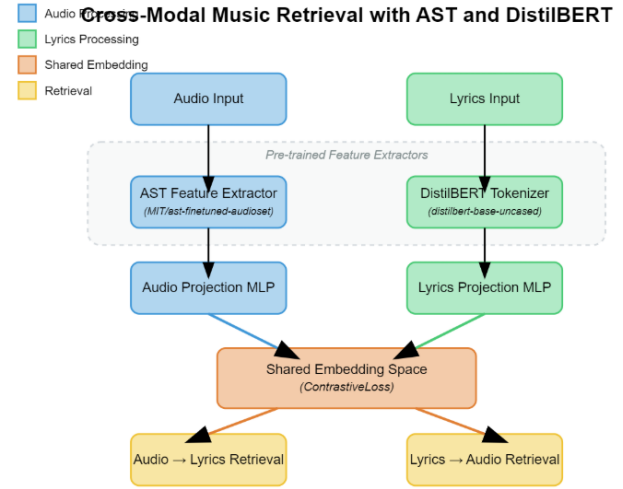


Figure 3: Architecture of Cross-Modal Retrieval.

CNN for audio and Doc2vec for lyrics within a Deep Canonical Correlation Analysis (DCCA) framework, we adopted DCCA as a sanity check on our fused representations.

We refined the architecture by fine-tuning pretrained models—AST for audio and DistilBERT for text—to learn a shared embedding space for paired music audio and lyrics. Only the last two layers of each pretrained model were unfrozen during training. We combined the PMemo and MERGE datasets to construct a set of approximately 2.5k aligned audio–lyrics pairs.

To optimize retrieval performance, we used the InfoNCE loss [11], which encourages the model to pull together embeddings of true audio–text pairs while pushing apart mismatched ones.

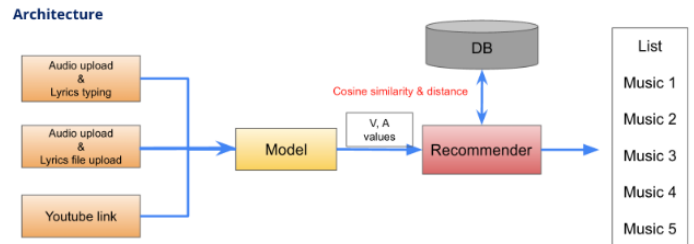


Figure 4: Architecture of Music Recommendation System.

3.6.2 Music Recommendation System. We also extended the system to serve as a music recommendation engine. Users can input either audio (via file upload) or lyrics (via direct input, file upload, or a YouTube link). The system predicts valence and arousal values for the input and recommends emotionally similar songs from a database.

Recommendations are generated using a hybrid similarity metric that combines cosine similarity and Euclidean distance to capture both directional and magnitude-based relationships in the valence-arousal space [12].

4 Results

4.1 Quantitative Results

4.1.1 Model Comparison Results. The results presented in the following tables provide a comprehensive evaluation of all Ablated Systems models used in this study. The VGGish-only model was excluded from the comparison due to its significantly inferior performance, with MSE and MAE values ranging between 0.3 and 0.4 and negative R^2 values, indicating that the model failed to learn effectively. These poor results would have made meaningful comparison with other models impractical.

Model	MSE	MAE
AST only	0.0213	0.1108
BERT only	0.0182	0.1026
AST + BERT	0.0146	0.0946
AST + BERT (modality)	0.0138	0.0879
AST + BERT (gated)	0.0149	0.0900
AST + BERT (Cross)	0.0131	0.0902

Table 1: MSE and MAE results on the MERGE Balanced dataset.

Model	V(CCC)	V(R^2)	A(CCC)	A(R^2)
AST only	0.5979	0.3595	0.6092	0.2196
BERT only	0.7107	0.4882	0.5728	0.2125
AST + BERT	0.7538	0.5854	0.6522	0.3737
AST + BERT (modality)	0.7629	0.5907	0.7094	0.4691
AST + BERT (gated)	0.7504	0.5557	0.6783	0.4427
AST + BERT (cross)	0.8035	0.6526	0.7202	0.4736

Table 2: Valence and Arousal prediction performance on the MERGE-Balanced dataset.

Model	Q1	Q2	Q3	Q4	Acc
AST only	0.7237	0.7338	0.4841	0.5000	0.6070
BERT only	0.6538	0.8000	0.7068	0.6211	0.6930
AST + BERT	0.7087	0.7040	0.6585	0.7174	0.6967
AST + BERT (modality)	0.8299	0.8684	0.7092	0.7500	0.7900
AST + BERT (gated)	0.8333	0.8456	0.6050	0.7386	0.7600
AST + BERT (cross)	0.8112	0.8859	0.7532	0.7143	0.7700

Table 3: F1 scores and Accuracy on MERGE-Balanced dataset.

4.1.2 Evaluation on Additional Datasets. This section evaluates our best-performing model—the AST+BERT with cross-attention multimodal system—when fine-tuned on different datasets. We tested this architecture using various training data configurations: *Merge-Balanced-Half* (using only half of the Merge-Balanced training set), *Merge-Complete* (which contains some class imbalance), and the *PMemo* dataset. These comparisons were made to observe how dataset characteristics impact the model’s performance in music emotion recognition tasks.

The results demonstrated that performance did not significantly decrease compared to our original baseline model. We were able to

achieve stable results even with reduced fine-tuning data. While Merge-Complete contained slightly more imbalance compared to Merge-Balanced, we observed that it performed better at predicting emotions in Quadrant 2 (Q2). However, this came at the cost of reduced performance in the remaining quadrants.

Dataset	MSE	MAE
Merge-Balanced-Half	0.0130	0.0863
Merge-Complete	0.0125	0.0801
PMemo	0.0108	0.815

Table 4: MSE and MAE of the AST+BERT (Cross) Model on Additional Datasets.

Dataset	CCC (V)	CCC (A)	R^2 (V)	R^2 (A)
Merge-Balanced-Half	0.7560	0.6162	0.7123	0.5004
Merge-Complete	0.7878	0.6281	0.7292	0.5151
PMemo	0.7705	0.5178	0.7985	0.6136

Table 5: Valence and Arousal Prediction Performance of the AST+BERT (Cross) Model on Additional Datasets.

Dataset	Q1	Q2	Q3	Q4	Acc
Merge-Balanced-Half	0.7612	0.7907	0.6753	0.6885	0.7200
Merge-Complete	0.7843	0.9286	0.7397	0.7229	0.7982
PMemo	0.8819	0.5000	0.3000	0.2667	0.7253

Table 6: F1 Scores and Accuracy of the AST+BERT (Cross) Model on Additional Datasets.

4.2 Baseline Model Performance

The baseline performance across different models shows encouraging results in predicting music emotions. From Table 1, we observe that our baseline AST+BERT model (simple concatenation fusion) achieves an MSE of 0.0146 and MAE of 0.0946. Looking at Table 2, this baseline model demonstrates good performance with a CCC of 0.7538 for valence and 0.6522 for arousal, along with R^2 values of 0.5854 and 0.3737, respectively.

Table 3 shows that the baseline model achieves reasonable F1 scores across quadrants with an overall accuracy of 0.6967. The AST+BERT (cross) model, which incorporates cross-attention, shows significant improvements across all metrics, highlighting the benefit of using more sophisticated fusion mechanisms.

4.3 Singlemodal Model Performance

The singlemodal results reveal insightful trends regarding the contributions of each modality. From Tables 1 and 2, we observe that the BERT-only model (using only lyrics) outperforms the AST-only model (using only audio) in terms of MSE (0.0182 vs. 0.0213) and MAE (0.1026 vs. 0.1108).

For specific emotion dimensions, Table 2 shows that the BERT-only model achieves better valence (positive/negative emotion)

prediction with higher concordance correlation coefficient (CCC) of 0.7107 compared to AST-only's 0.5979. However, the AST-only model slightly outperforms in arousal (intensity) prediction with a CCC of 0.6092 versus BERT-only's 0.5728. This suggests that lyrics tend to better capture the positivity/negativity of emotions, while audio features better capture emotional intensity. Similar findings were reported in prior work using multilingual transformer models to predict valence and arousal from text [13]

4.4 Multimodal Model Performance

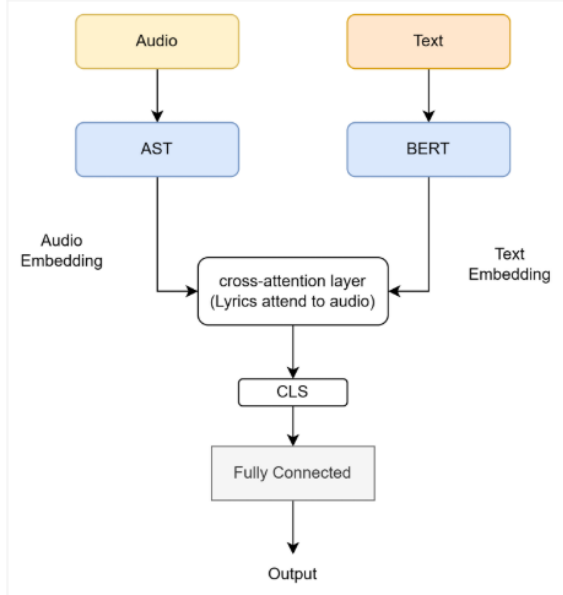


Figure 5: Architecture of the Cross-Attention Multimodal Model.

The multimodal approaches consistently outperform singlemodal approaches across all metrics. As shown in Table 3, all multimodal models achieve higher F1 scores and accuracy in quadrant-based classification.

The AST+BERT (cross) model demonstrates the best overall performance with:

- Highest valence and arousal prediction (CCC: 0.8035 and 0.7202)
- Highest R^2 scores (valence: 0.6526, arousal: 0.4736)
- Best quadrant performance in Q2 (0.8859) and Q3 (0.7532)
- Second-highest overall accuracy (0.7700)

The AST+BERT (modality attention) model performs best in Q4 (0.7500) and achieves the highest overall accuracy (0.7900). Meanwhile, the AST+BERT (gated fusion) model excels in Q1 (0.8333).

These results clearly demonstrate that cross-attention mechanisms effectively integrate audio and lyrical information, capturing complementary emotional cues from both modalities and yielding superior music emotion recognition performance.

4.5 Qualitative Analysis

4.5.1 Human Annotator Results for 10 Modern Songs. To evaluate our model's performance on contemporary music, we collected

Song-ID	Annotator 1			Annotator 2			Annotator 3		
	V	A	Q	V	A	Q	V	A	Q
1	0.45	0.70	Q2	0.30	0.60	Q2	0.40	0.55	Q2
2	0.90	0.80	Q1	0.85	0.60	Q1	0.80	0.80	Q1
3	0.20	0.70	Q2	0.30	0.90	Q1	0.40	0.85	Q1
4	0.80	0.90	Q1	0.90	0.85	Q1	0.90	0.90	Q1
5	0.20	0.30	Q1	0.40	0.45	Q3	0.60	0.40	Q4
6	0.70	0.65	Q3	0.70	0.70	Q3	0.60	0.60	Q3
7	0.25	0.45	Q3	0.20	0.40	Q2	0.40	0.60	Q4
8	0.35	0.90	Q3	0.30	0.60	Q2	0.60	0.60	Q2
9	0.90	0.90	Q1	0.85	0.70	Q1	0.90	0.90	Q1
10	1.00	1.00	Q1	0.85	0.90	Q1	0.95	0.90	Q1

Table 7: Valence, arousal, and quadrant ratings by individual annotators for modern songs.

annotations from three human raters for 10 modern songs released between 2000–2025. Human annotators rated valence and arousal on continuous scales (0–1) based on predefined intensity levels and emotional examples provided in the annotation guidelines (see Appendix C). Table 7 presents the individual ratings from each annotator, including valence, arousal, and quadrant assignments.

The annotation process revealed several interesting patterns. First, we observed substantial inter-annotator agreement for songs with extreme emotional characteristics (e.g., Songs 4, 9, and 10), which all annotators consistently placed in Q1 (high valence, high arousal). However, there was notable disagreement for songs with more ambiguous emotional content, particularly Songs 5, 7, and 8, where annotators assigned different quadrants.

These variations highlight the inherently subjective nature of emotional perception in music. For instance, Song 5 received quadrant assignments ranging from Q1 to Q4, suggesting that different listeners may focus on distinct musical or lyrical elements when forming emotional judgments. Notably, we observed that human listeners tend to gravitate toward the more definitive emotional elements when confronted with ambiguous audio or lyrical content, rather than attempting to precisely balance conflicting cues as our computational models do. This natural human tendency to resolve emotional ambiguity by leaning toward the more salient emotional signals presents a significant challenge for computational models attempting to predict consensus emotional responses.

4.5.2 Comparison between Human and Model Results. Table 8 presents a comprehensive comparison between the averaged human annotations and predictions from our three multimodal models (cross-attention, gated fusion, and modality attention), alongside predictions from GPT-4o as an external reference point.

Our analysis reveals that the cross-attention model achieved the closest alignment with human judgments for valence prediction, particularly for songs with high consensus among annotators (e.g., Songs 2, 9, and 10). The gated fusion model demonstrated strong performance in high-arousal songs but tended to underestimate arousal levels in more subdued compositions. The modality attention model excelled at detecting subtle emotional nuances in songs with contrasting audio and lyrical cues.

Figure 6 visualizes these comparisons in valence–arousal space for each song. The visualization reveals that while our models

Song-ID	Annotator Avg			Cross Attention			Gated Fusion			Modality Attention			GPT-4o		
	V	A	Q	V	A	Q	V	A	Q	V	A	Q	V	A	Q
1	0.383	0.617	Q2	0.366	0.538	Q2	0.416	0.635	Q2	0.391	0.557	Q2	0.650	0.450	Q4
2	0.733	0.600	Q1	0.657	0.549	Q1	0.756	0.596	Q1	0.846	0.629	Q1	0.900	0.650	Q1
3	0.783	0.600	Q1	0.487	0.293	Q2	0.598	0.316	Q2	0.525	0.362	Q2	0.900	0.450	Q2
4	0.867	0.600	Q1	0.296	0.580	Q2	0.621	0.373	Q2	0.594	0.366	Q2	0.900	0.450	Q2
5	0.383	0.400	Q4	0.436	0.546	Q3	0.436	0.360	Q4	0.405	0.481	Q3	0.400	0.550	Q3
6	0.733	0.650	Q3	0.625	0.492	Q4	0.662	0.515	Q4	0.653	0.535	Q4	0.850	0.600	Q1
7	0.283	0.483	Q3	0.373	0.239	Q4	0.424	0.412	Q3	0.493	0.376	Q3	0.400	0.650	Q3
8	0.417	0.600	Q2	0.553	0.422	Q4	0.424	0.412	Q3	0.584	0.383	Q3	0.550	0.550	Q2
9	0.883	0.767	Q1	0.738	0.683	Q1	0.552	0.649	Q1	0.655	0.636	Q1	0.900	0.900	Q1
10	0.933	0.933	Q1	0.694	0.699	Q1	0.854	0.498	Q1	0.802	0.622	Q1	0.900	0.900	Q1

Table 8: Comparison of valence, arousal, and quadrant predictions across AST+BERT (cross), AST+BERT (gated), AST+BERT (modality attention), GPT-4o models, and annotator averages.

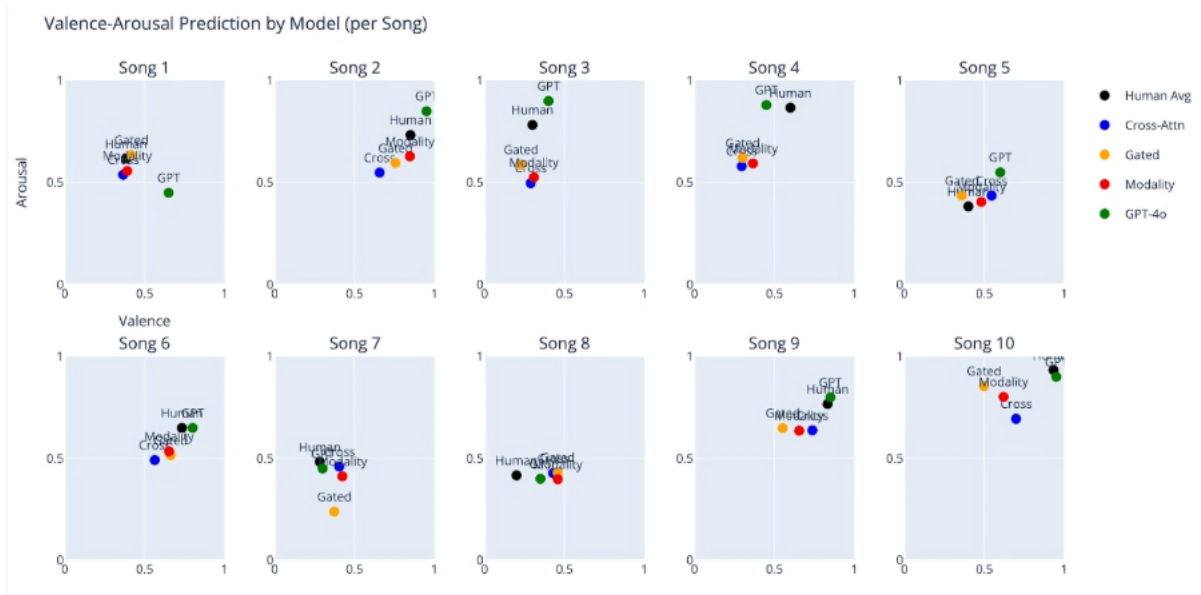


Figure 6: Valence–Arousal scatter plots for each song, comparing predictions by human annotators, cross-attention model, gated fusion model, modality attention model, and GPT-4o.

generally cluster near human annotations, there are systematic differences in how they position songs in the emotional space. For example, the cross-attention model tends to predict more moderate arousal values than human annotators for high-arousal songs (Songs 1, 2, and 10), suggesting potential room for calibration.

4.5.3 Error Analysis. Our qualitative analysis identified two representative cases where emotional prediction proved particularly challenging for our models, offering valuable insights into the limitations of current approaches.

Case 1: Difficulty Interpreting Metaphorical Expressions. Song 4 presents a complex case where metaphorical language creates a significant gap between literal meaning and emotional intent. As shown in Figure 6, human annotators consistently placed this song in the high-valence, high-arousal quadrant (Q1), while our

models (cross-attention, gated fusion, and modality attention) positioned it in the lower-arousal range of Q2.

The lyrics contain numerous negative phrases (e.g., “ash and dust,” “sweat my rust,” “breathing in the chemicals,” “breaking in,” “prison bus,” “apocalypse”), which likely influenced our models toward a negative valence prediction. However, human listeners interpreted these metaphorical expressions within the context of the song’s energetic audio elements and overall theme of transformation, resulting in a more positive emotional assessment.

This case highlights a crucial limitation: current NLP approaches like BERT primarily process text at a semantic level without fully capturing the nuanced metaphorical meanings that human listeners intuitively understand within musical contexts.

Case 2: Ambiguity in Emotional Expression. Song 5 exemplifies the challenge of songs with inherently ambiguous emotional content. The scatter plot reveals wide disagreement not only between human annotators and models but among the models themselves. Most notably, human annotators placed this song in the moderate arousal range, while GPT-4o predicted substantially higher arousal.

The lyrics contain numerous negative expressions at the sentence level (e.g., “not proud,” “torn up town,” “we don’t care,” “never be royals”), yet the overall message can be interpreted in both positive and negative ways. The song juxtaposes rejection of luxury (“Cristal, Maybach, diamonds”) with an underlying message of contentment with simpler life, creating emotional ambiguity that different listeners may resolve differently.

This ambiguity explains the significant variance in quadrant assignments among human annotators (ranging from Q1 to Q4) and highlights a fundamental challenge: when emotional signals are mixed, humans tend to gravitate toward either the audio or lyrical cues that seem more definitive, rather than attempting to precisely balance conflicting signals as computational models do.

Both cases underscore the need for more sophisticated semantic understanding in multimodal music emotion recognition—particularly for capturing metaphorical language, cultural context, and resolving emotional ambiguity in ways that align with human perception patterns.

4.6 Deep Canonical Correlation Analysis

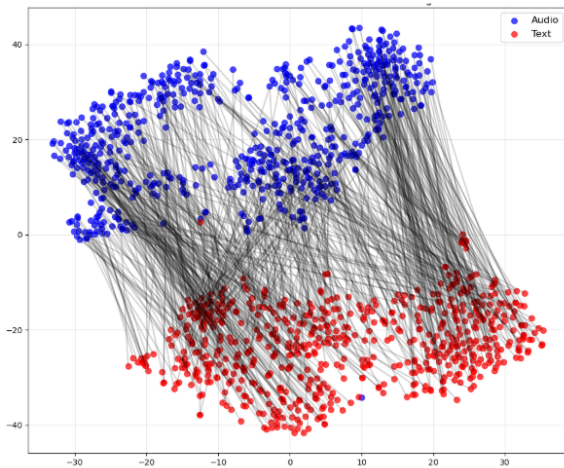


Figure 7: t-SNE visualization of audio and text embeddings. Each line connects paired audio and text features, with blue representing audio embeddings and red representing text embeddings.

We visualized the embedding space using t-SNE, as shown in Figure 7. Each black line connects a paired audio and text embedding. The clear separation between the two modalities highlights the need for effective cross-modal alignment, which our contrastive loss was designed to address.

The DCCA results further demonstrate that our model has effectively learned a shared feature space in which corresponding audio and lyric embeddings are brought closer together while still



Figure 8: Visualization of Recall@K and ranking metrics for cross-modal retrieval tasks. Blue bars represent Audio → Lyrics retrieval performance, while green bars represent Lyrics → Audio retrieval.

preserving the distinct structure of each modality. Figure 7 displays the relationship between audio embeddings (blue points) and lyric embeddings (red points). The gray lines connecting paired inputs indicate that the model successfully captures semantic relationships between modalities. While the two clusters remain modality-specific, the emergence of partially overlapping regions suggests that the model is learning meaningful cross-modal correspondences without collapsing modality-specific information.

In Figure 8, we report Recall@K performance for both Audio → Lyrics and Lyrics → Audio retrieval tasks. At Recall@1, performance reaches approximately 23–24%, improves to 47% at Recall@5, and exceeds 56% at Recall@10. Ranking metrics further reveal that while median rank remains low (7) across both directions, the mean rank is influenced by several harder samples. Although these retrieval outcomes are not ideal, they reflect promising alignment quality and justify further use of these features in downstream tasks such as emotion prediction and music recommendation.

4.7 Recommendation System

To demonstrate the practical utility of our emotion prediction model, we implemented a recommendation system that suggests emotionally similar songs based on valence and arousal (V, A) values. The system takes user input through either uploaded audio–lyric files or YouTube links and predicts the emotional vector using our best-performing AST+BERT (cross-attention) model.

We initially applied cosine similarity to retrieve similar songs, but the results occasionally failed to align with human perception. For instance, a test song with predicted values of (V=0.51, A=0.49) returned distant matches such as (0.30, 0.30) or (0.70, 0.70), while skipping songs with nearly identical V and A values like (0.50,

0.49). This mismatch stems from cosine similarity focusing solely on vector direction without accounting for absolute distance.

To overcome this limitation, we introduced a hybrid similarity metric that combines cosine similarity and Euclidean distance. This approach is inspired by previous research on hybrid similarity learning in representation spaces [12]. Specifically, we compute the final similarity score as a weighted combination of cosine similarity and inverse Euclidean distance:

$$\text{HybridSim} = \alpha \cdot \text{cosine_sim} + (1 - \alpha) \cdot (1 - \text{euclidean_dist}), \quad (3)$$

where α was empirically set to 0.7. This weighting prioritizes directional alignment while still accounting for magnitude proximity.

The examples shown in Table 9 illustrate how this hybrid method leads to more emotionally consistent recommendations. Informal listening evaluations confirmed that the retrieved songs were perceptually aligned with the emotional tone of the query, reinforcing the utility of this system for emotion-aware music recommendation tasks.

	Valence	Arousal		Valence	Arousal
Test 1	0.50	0.49	Test 2	0.40	0.53
Song1	0.48	0.46	Song1	0.40	0.53
Song2	0.49	0.45	Song2	0.41	0.53
Song3	0.49	0.45	Song3	0.40	0.54
Song4	0.48	0.45	Song4	0.40	0.54
Song5	0.47	0.45	Song5	0.40	0.54

Table 9: Top-5 closest songs Recommended by the Music Recommendation system for Test songs 1 and 2 Based on Valence-Arousal Similarity

5 Conclusion

5.1 Summary

This project explored a multimodal approach to music emotion recognition by combining audio and lyrics features[14] using pre-trained AST and BERT models. Framing the task as a regression problem in the Valence–Arousal space, we fine-tuned models on the MERGE and PMEmo datasets and evaluated them using both regression and classification metrics.

5.2 Key Findings

Our results show that multimodal fusion models significantly outperform single-modal models in music emotion recognition. Among the tested architectures, the cross-attention model consistently achieved the highest performance across all evaluation metrics, including MSE, MAE, CCC, and quadrant-level F1 scores.

An analysis of modality-specific contributions revealed that lyrics predominantly contributed to accurate valence predictions, while audio features provide stronger cues for arousal prediction. This underscores the complementary nature of the two modalities.

Moreover, the cross-attention mechanism was particularly effective in capturing and integrating emotional cues from both modalities, demonstrating robust generalization even under limited data conditions. Although the training dataset primarily consisted of

older songs, evaluations on modern songs, supported by human annotations, further confirmed that the model generalized effectively to contemporary musical styles, particularly in cases with high annotator consensus.

5.3 Insights

The findings of this study suggest that multimodal models are better equipped to capture the complex and nuanced nature of musical emotions, especially when lyrics and audio either reinforce or contrast with one another emotionally. In particular, our cross-attention mechanism effectively integrated emotional signals from both modalities, rather than simply averaging conflicting cues.

However, human annotators tend to resolve ambiguity by focusing on the most dominant emotional cue—be it in the lyrics or the music. This behavioral difference highlights a fundamental challenge in replicating human-like emotional reasoning. Additionally, many of the observed prediction errors were linked to metaphorical language or ambiguous lyrical content, which current language models like BERT still struggle to interpret in musically relevant ways.

5.4 Limitations

Despite the strong performance of our proposed models, several limitations remain. Current models lack the ability to deeply interpret figurative language, cultural references, or nuanced emotional expressions in lyrics, which can lead to misinterpretations.

Furthermore, while our models have been evaluated on popular modern songs, their ability to generalize to lesser-known or niche tracks—which may exhibit substantially different musical characteristics—has not yet been tested.

The datasets primarily contain songs from Western or mainstream music traditions, predominantly in English. This lack of cultural and linguistic diversity may limit the generalizability and applicability of our model to more diverse global music contexts, suggesting a clear area for future improvement.

Lastly, the architecture assumes an equal emotional contribution from audio and lyrics for every song, which may not hold across all genres or musical styles where one modality could dominate the emotional expression.

5.5 Future Work

5.5.1 Applying a Higher Penalty to Valence MSE. Our findings indicate that valence is more challenging to predict accurately than arousal. Future training strategies could explore asymmetric loss functions that weight valence errors more heavily, to guide the model toward improved performance in emotional positivity estimation.

5.5.2 Enhancing Recommendation System Diversity Using Genre Information. To improve the diversity and personalization of our recommendation system, we plan to incorporate genre metadata. Emotionally similar songs from distinct genres can create more engaging and varied listening experiences, and genre-aware filtering may also help reduce redundancy in recommendations.

5.5.3 Enhancing Multimodal Classification with DCCA. Inspired by Sun’s work [15], we will adapt DCCA to learn compact projections

that capture shared information across audio and text while filtering out modality-specific noise.

Appendix

A Q&A

Q1. In a music recommendation system, recommendation should involve not only finding similar songs but also introducing a level of creativity by suggesting different songs each time. How would you address this aspect?

A1. To incorporate creativity into the recommendation system, we plan to implement a controlled randomness mechanism. Specifically, the system could randomly select among the top-10 ranked songs to maintain emotional relevance while ensuring diversity. This would allow the system to prioritize emotionally similar songs while still introducing variability in the recommendations. Additionally, incorporating factors such as genre diversity or lyrical theme diversity could further enhance the creative aspect of the recommendations.

Q2. How could your model be improved with newer architectures (e.g., transformers with self-attention)?

A2. Our model already uses transformer-based architectures like AST and BERT, both built on self-attention. We applied fusion methods like cross-attention and found it to perform relatively better across most metrics. This shows that self-attention helps effectively integrate emotional signals from audio and lyrics. Future work could explore newer transformers like Longformer to handle longer lyrics or train multimodal models end-to-end.

Q3. How did you handle the dataset bias toward older music?

A3. Recognizing that our training dataset primarily consists of older songs, especially from English-speaking Western countries, we explicitly evaluated our model’s performance using a selection of contemporary songs (2000–2025) to assess generalization capabilities. However, we did not apply explicit debiasing methods during training. Future work should consider employing strategies such as weighted sampling, targeted data augmentation, or adversarial debiasing methods to better mitigate the bias toward older and Western music, and to improve generalization across different eras and cultural backgrounds.

Q4. Would including more diverse or multilingual lyrics help?

A4. Yes. Incorporating diverse and multilingual lyrics could improve the model’s robustness and reduce cultural bias. Since emotions are expressed differently across languages and cultures, expanding the training data could enable the model to better capture nuanced emotional patterns and generalize to global music. This is especially relevant as the current dataset (MERGE) is primarily composed of English songsDeepLearner mod.

Q5. Could metaphor detection improve the model’s handling of ambiguous lyrics?

A5. Yes. One key error case in our analysis involved Song 4 (Radioactive), where metaphorical phrases were misinterpreted by the

model, while human raters correctly inferred the intended emotion. This suggests that adding a metaphor detection module—or fine-tuning with datasets that include metaphorical labels—could significantly improve valence prediction, especially when lyrics are emotionally ambiguous or metaphorical.

B List of 10 Modern Songs Used for Human Annotation

The information for Songs 1–10 used in the evaluation for Section 4.4 (Results) is provided below.

No.	Artist	Title
1	Oasis	Don’t Look Back in Anger
2	Pharrell Williams	Happy
3	Eminem	Lose Yourself
4	Imagine Dragons	Radioactive
5	Lorde	Royals
6	Ed Sheeran	Shape of You
7	Adele	Someone Like You
8	Sam Smith	Stay With Me
9	Shakira	Try Everything
10	Bruno Mars	Uptown Funk

Table 10: List of the 10 modern songs selected for human annotation, including artist names and song titles.

No.	YouTube Link	Time Range
1	https://www.youtube.com/watch?v=r8OipmKfPDcM	1:07–1:27
2	https://www.youtube.com/watch?v=ZbZSe6NBXs	0:00–0:20
3	https://www.youtube.com/watch?v=hyp-nX2s	0:48–1:08
4	https://www.youtube.com/watch?v=ktvTqknDobU	1:14–1:34
5	https://www.youtube.com/watch?v=nlcIKh6sBtc	0:16–0:36
6	https://www.youtube.com/watch?v=JGwWNGJdvx8	0:30–0:50
7	https://www.youtube.com/watch?v=hLQl3WQQoQ0	0:57–1:17
8	https://www.youtube.com/watch?v=pB-5XG-DdAA	0:41–1:01
9	https://www.youtube.com/watch?v=c6rP-YP4c5I	0:49–1:09
10	https://www.youtube.com/watch?v=OPf0YbXqDm0	0:54–1:14

Table 11: YouTube links and corresponding time ranges of the 10 modern songs used for human annotation.

C Human Annotation Guidelines

Valence	Intensity	Example Emotions
0.0	Extremely Negative	Despair, Hatred, Terror
0.2	Strongly Negative	Sadness, Depression, Anger
0.4	Mildly Negative	Disappointment, Longing, Loneliness
0.5	Neutral	Indifference, Confusion, Ambiguity
0.6	Mildly Positive	Hope, Calmness, Feeling Good
0.8	Strongly Positive	Joy, Gratitude, Love
1.0	Extremely Positive	Euphoria, Celebration

Table 12: Valence Annotation Guidelines

Arousal	Intensity	Example Emotions
0.0	Extremely Calm	Sleepiness, Meditation, Lethargy
0.2	Low Arousal	Tranquility, Peacefulness
0.4	Below Moderate	Quiet Sadness, Low Energy
0.5	Moderate	Subtle Sadness, Focus
0.6	Above Moderate	Excitement, Surprise, Tension
0.8	High Arousal	Anger, Thrill, Intense Joy
1.0	Extremely Aroused	Panic, Mania, Frenzy

Table 13: Arousal Annotation Guidelines

D Code

<https://github.com/wldnjs95/Multimodal-Music-Emotion-Recognition>

References

- [1] J. Lee, S. Choi, T. Moon, M. Bong, and J. C. Lee. Merge: Multilingual emotional response generation evaluation dataset (version 1.0). <https://doi.org/10.5281/zenodo.13939205>, 2024. Dataset.
- [2] J. Lee, S. Choi, T. Moon, M. Bong, and J. C. Lee. Merge: Multilingual emotional response generation evaluation dataset. *arXiv preprint arXiv:2407.06060*, 2024.
- [3] J. A. Russell. A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6):1161–1178, 1980.
- [4] K. Zhang, H. Zhang, S. Li, C. Yang, and L. Sun. The pmemo dataset for music emotion recognition. In *Proceedings of the 2018 ACM on International Conference on Multimedia Retrieval (ICMR '18)*, pages 135–142. ACM, 2018.
- [5] H. Zhang. Pmemo: Dataset for music emotion recognition. <https://github.com/HuiZhangDB/PMemo>, n.d. Computer software.
- [6] Y. Gong, Y.-A. Chung, and J. Glass. Ast: Audio spectrogram transformer. *Inter-speech*, pages 571–575, 2021.
- [7] MIT CSAIL. Mit/ast-finetuned-audioset-10-10-0.4593. <https://huggingface.co/MIT/ast-finetuned-audioset-10-10-0.4593>, n.d. Pretrained model.
- [8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019*, pages 4171–4186, 2019.
- [9] Google. bert-base-uncased. <https://huggingface.co/bert-base-uncased>, n.d. Pre-trained model.
- [10] Y. Yu, J. Tang, Y. Xiang, and X. Liu. Deep cross-modal correlation learning for audio and lyrics in music retrieval. *ACM Transactions on Multimedia Computing, Communications, and Applications*, 15(1):1–16, 2019.
- [11] G. Andrew, R. Arora, J. Bilmes, and K. Livescu. Deep canonical correlation analysis. In *Proceedings of the 30th International Conference on Machine Learning (ICML)*. PMLR, 2013.
- [12] J. Zhu, H. Zeng, S. Liao, Z. Lei, C. Cai, and L. Zheng. Deep hybrid similarity learning for person re-identification. *arXiv*, 2017.
- [13] G. A. Mendes and B. Martins. Quantifying valence and arousal in text with multilingual pre-trained transformers. In *European Conference on Information Retrieval*. Springer Nature Switzerland, 2023.
- [14] A. Jamdar, D. K. Vishwakarma, and P. Mishra. Emotion analysis of songs based on lyrical and audio features. *arXiv preprint arXiv:1506.05012*, 2015.
- [15] Z. Sun, C. Wang, G. Zhou, and Y. He. Multi-modal sentiment analysis using deep canonical correlation analysis. *arXiv preprint arXiv:1907.08696*, 2019.